



LightCOM

Lighthouse Consortium on
Obesity Management

Study name: **Lighthouse Consortium on Obesity Management**
Short name: **LightCOM**

SOP title: **Standard Operating Procedure for the use of SENS
motion in LightCARE and LightWAY**

SOP short title: **SOP on SENS**

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1 PURPOSE AND SCOPE

This document outlines the procedure for using the SENSmotion® activity sensor to measure the participant's physical activity in the LightCARE, LightWAY and LightBAR trials.

2 PERSONNEL QUALIFICATIONS AND RESPONSIBILITIES

- Training in using the SENSmotion® app and the SENS web app
- Training in assembling and installing the SENS patch

This SOP should be referred to for training purposes. An experienced staff member should conduct all practical training.

3 EQUIPMENT AND RESOURCES

3.1 Equipment

- SENS sensor
- SENS patch
- Disinfection cloth
- Computer and access to the SENS web app
- Investigator mobile phone or tablet with the SENS motion app (not participant's phone)

3.2 Forms

- Information relating to SENS will be collected in the eCRF in OpenClinica.
 - Did the participant receive the SENS motion sensor?
 - What is the SENSmotion® sensor number?
 - Start time of SENS measurements
 - End of SENS measurements
 - Has the participant returned the SENSmotion® sensor?
 - Has the SENS data been uploaded?

3.3 Other Resources

- Envelope
- Prepaid bubble envelope
- Information letter to participants

4 HEALTH AND SAFETY

- NA

5 PROCEDURE

1. Access to SENS web app
2. Pre-package return envelopes
3. Download the SENS motion app
4. Turning on the SENS motion sensor
5. Register a user in the SENS system and assign a sensor
6. Assign a sensor to registered user
7. Placing the SENS motion sensor on the participant

8. Receiving the SENS motion sensor
9. Reminders

5.1 Access to SENS web app

- Write to contact@sens.dk and request access to the web app at <https://app.sens.dk/>, specifying the project and email address to be used for your account.

5.2 Pre-package return envelopes

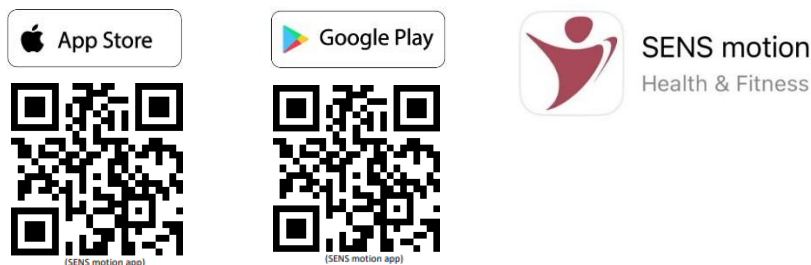
- Take an envelope for pre-packing and include the following:

Information letter (CID)

- a. A prepaid bubble envelope
 - b. A spare patch
 - c. A disinfection cloth
- Place the envelope in the cardboard box marked "Pre-packaged envelopes."

5.3 Download the SENS motion app

- Search for "SENS motion" in the App Store or Google Play or scan the QR code.



5.4 Register a user in the SENS system and assign a sensor

To register a user, a sensor needs to be assigned.

- **Step 1:** The following needs to be located
 - a) participant's ID number in LightCOM (eCRF in OpenClinica)
 - b) The SENS motion sensor the participant will carry for the next 7 days
- **Step 2:** Log on to the SENS web app (<https://app.sens.dk/r/login>). Then click on "Projects" and choose the trial the participant belongs to: "LightCARE", "LightWAY", or "LightBAR".

Organisations overview

Sort by Name ⌵ ⌴ 🔍 Search by Name...			
Name	Organisation Type	Projects	Actions
LightCOM	Standard	6	📄 Projects

- **Step 3:** Select "Add participant" in Participants overview

LightWAY / Participants

Participants overview

In progress 21 Not started yet Completed 363 All 384

Sort by Creation date   Search by name or sensor name

Add participant Bulk export 

- **Step 4:** In the "Participant ID/First Name" box type the participant's ID nr. in LightCARE, LightWAY, or LightBAR followed by _and CID number (example: DK10_1024_CID1 for the first CID and DK10_1024_CID3 for the third CID), then select the relevant time zone.

NOTE: Do not add the participant first name in this box

Add Participant



Participant ID/First name * Last Name

Rasmus Jensen

Phone Number Timezone *

+45  88776655 (GMT+01:00) Brussels, Copenhagen 

- **Step 5:** Select "Activity & Sleep" under "Measurement Method"
- **Step 6:** Choose the sensor you have located from the drop-down menu under "Sensor on Thigh", the sensor ID is written on its label. You can search for the sensor by typing in the ID number. Then, in the eCRF in OpenClinica, register the sensor ID number in the "Physical Function" case report form in the CID section. First select "Yes" under "Did the participant receive the SENS sensor?" (PF09), then add the ID number next to "What is the SENS number?" (PF09a).

Measurement Method *

SENS Activity & Sleep (Newest) (v5.5) 

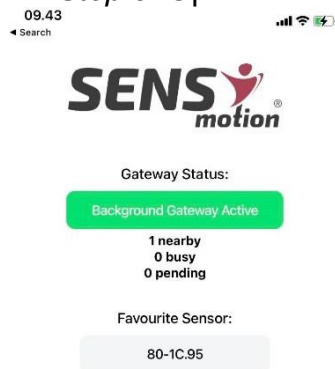
Sensor on Thigh *  Select a sensor 

Search sensor...

- **Step 2:** In “Sensor details” click “Actions” and “Turn sensor on”

The screenshot shows the SENS web interface. At the top, there's a navigation bar with 'SENS' logo, 'Workspace', 'Sensors', and 'Files' tabs. Below this is a green banner with a welcome message and a link to release notes. The main content area is titled 'Sensor details' with a status 'Off'. It displays four columns of information: Name (I-B6C.37F), Battery Level (Est. 74 days left), Data on Sensor (0 days used of 10), and Pending Action (No pending action). On the right, an 'Actions' menu is open, showing options: 'Move sensor', 'Turn sensor on' (highlighted with a red box), 'Turn sensor off', and 'Cancel pending'.

- **Step 3:** Open the SENSmotion® app on the phone or tablet close to the sensor/s.



NOTE: Shake the sensor a few times to activate the accelerometer in the sensor.

NOTE: The sensor only turns on and tracks activity when the status changes to "On."

5.6 Placing the SENS motion sensor on the participant

First, the sensor and the patch are assembled and then placed on the patient's leg.

5.5.1 Assemble the sensor and the patch



- **Step 1:** You will need:
 - a) a disinfection cloth
 - b) a sensor
 - c) a patch



- **Step 2:** Place the sensor in the patch pocket with the SENSmotion® logo facing upward and the sensor label facing down to the widest part of the patch.



- **Step 3:** Pull down the bottom piece (tab 1) of foil/liner to allow the sensor and patch to be pressed together firmly



- **Step 4:** Press the patch thoroughly around the edges of the sensor to secure it.

5.5.2 Installation of the sensor patch on the patient's leg

The patch should be placed on the OUTER SIDE of the thigh above the patient's knee. The round part of the sensor should point DOWN toward the knee, and the upper part should point TOWARD the hip.



- *Step 1:* Clean the skin where the patch is meant to be placed.



- *Step 2:* Peel off the paper on the back of the patch using tab 2



- *Step 3:* Gently place the patch on the outer side of the thigh. Press firmly to ensure it adheres evenly.



- *Step 4:* If your patch has a white edge (tab 3), remove it. If not, the sensor application is complete after step 3.

- *Step 5:* Give the participant the prepacked envelope with instructions and an extra patch.

5.7 Receiving the SENS Sensor

- *Step 1:* Remove the sensor from the patch if it's still attached, being careful not to remove the ID number on the sensor. Clean the sensor with a disinfection cloth. Dispose of the patch in the bin.
- *Step 2:* Open the SENSmotion® app and place the phone/tablet near the received sensor/s. The sensor/s and the app will now detect each other, and data will be uploaded.



Gateway Status:

Background Gateway Active

1 nearby
1 busy
0 pending

Favourite Sensor:

80-1C.95

Syncing 96%

- **Step 3:** Log on to the SENS web app (<https://app.sens.dk/r/login>). Then click on "Projects" and choose the trial the participant belongs to: "LightCARE", "LightWAY", or "LightBAR".

Organisations overview

Sort by Name ↑ Search by Name...			
Name	Organisation Type	Projects	Actions
LightCOM	Standard	6	Projects

- **Step 4:** In the "Completed" tab find and click on the sensor info matching the received sensor

[LightWAY](#) / [Participants](#)

Participants overview

In progress 21	Not started yet	Completed 363	All 384	⋮
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In progress 21		Not started yet		Completed 363	All 584		
Sort by Creation date		Search by name or sensor name		Add participant		Bulk export	
	Name	Start Time	End Time	Sensor Info	Status	Actions	
	DK07_1056_CID2	09.01.2026	16.01.2026	I-3E3.0C1 Thigh	Completed		
	DK07_1057_CID2	12.01.2026	19.01.2026	I-EA6.830 Thigh	Completed		
	DK07_1128_CID1	12.01.2026	19.01.2026	I-9BC.C05 Thigh	Completed		
	W.YS08_1011.00	14.01.2026	21.01.2026	E-3FB.071 Thigh	Completed		
	DK06_1118_CID1	14.01.2026	21.01.2026	I-4A9.FBF Thigh	Completed		
	DK07_1070_CID2	14.01.2026	21.01.2026	I-754.E0E Thigh	Completed		

Step 5: In "Sensor details" press "Actions" for the sensor you want to turn off and then press "Turn sensor off."

Sensor details On

Name
I-754.E0E

Battery Level
Battery: Est. 21 days left

Data on Sensor

 Est. 7 days used (of 10)

Pending Action
No pending action

Actions

Actions

- Move sensor
- Turn sensor on
- Turn sensor off**
- Cancel pending

- Step 6:** When the status for the selected sensor/s has changed to "Off," in the "physical function" case report form in the eCRF in OpenClinica record that the SENS motion sensor has been returned by selecting "Yes" under "Have the participant returned the SENS sensor?" (PF09e) and that data has been uploaded by selecting "Yes" under "Is the data from the sensor uploaded?" (PF09d).

NOTE: Make sure that the status has changed to "Off" before you move the sensor/s away from the phone/tablet.

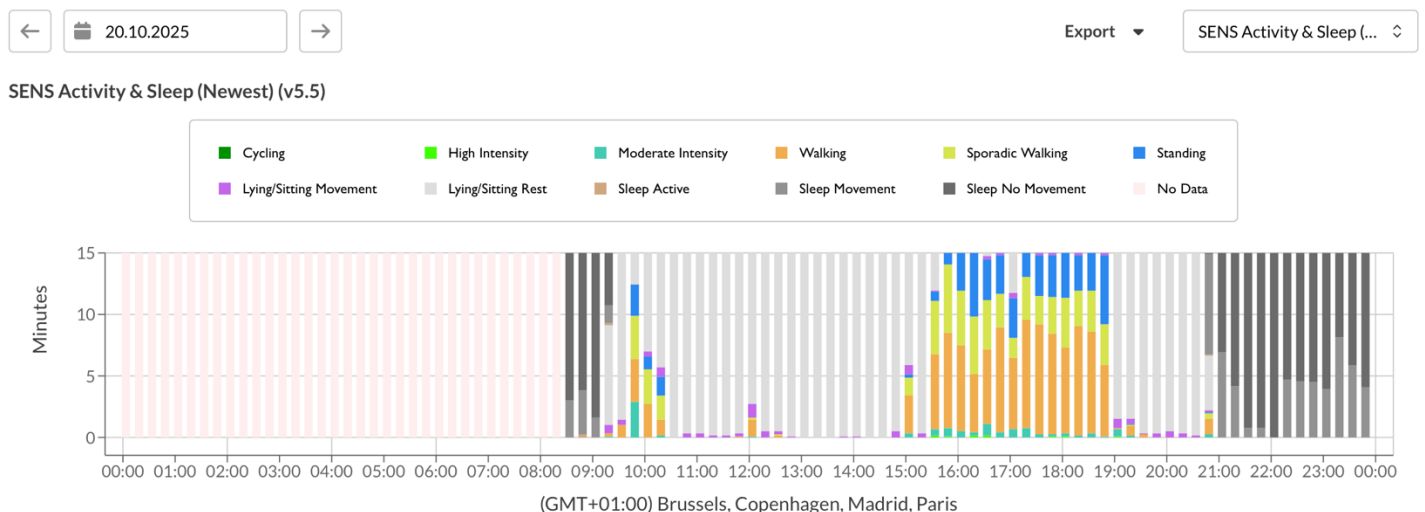
- **Step 7:** If the sensors battery has ≥ 2700 mV remaining, place the sensor in the "Sensors" box, but if the sensors battery has < 2700 mV remaining, place the sensor in the "Used Sensors" box.

5.8 Verifying and cleaning data

- **Step 1:** Verify that the data looks correct (Video 1).
Using arrows next to the date above Activity (v5.5), look through the 7 days and see if the data looks legit. Things to look for:
 - Empty (light grey) spaces
 - One color e.g., blue or green, that is filling up a lot of the time.

If you find something suspicious or if you are unsure, look at the temperature and the accelerometer. In the top right corner, where it says Activity (v5.5) select:

- "Temperature" and see if there is a drop in the temperature. This indicates that the sensor is not attached to the participant. Don't mind the actual temperature; it is inaccurate, but changes indicate that the sensor has changed location.
- "Accelerometer" and look at the red line; it should be between 0 and -1. If the red line is between 0 and 1, the sensor is upside down, so it is either not attached to the participant or attached upside down. If you suspect the sensor is attached upside down, note this as a Protocol deviation in Open Clinica; this can be fixed.
- **Step 2:** Clean the data if data is not complete for the 7 days.
The data could be defective either a) at the start/end of the period or b) in the middle of the period. If it is:
 - a) Click on Edit Patient and adjust the Monitor period according to the clean data (i.e., select a specific time for when data is legit). This should be noted as a Protocol deviation in Open Clinica.
 - b) Click on Edit Patient, and in the box "Last Name", write data incomplete see protocol deviation. Note this as a Protocol deviation in Open Clinica; write the period/s for when data is defective/missing.
- **Step 3: Remember that** if data is not complete for the 7 days, this should be noted in Protocol deviations in Open Clinica.



6. EXPECTED RESULTS, TROUBLESHOOTING AND CORRECTIVE ACTION

- NA

7. DATA AND RECORDS MANAGEMENT

- In the eCRF in OpenClinica the following will be registered:
 - Did the participant receive the SENS motion sensor?
 - What is the SENS motion sensor number?
 - Start time of SENS measure
 - End of SENS measure
 - Has the participant returned the SENS motion sensor?
 - Is the SENS data uploaded?

8. REFERENCE

- SENS motion® For Research: User Manual

9. HISTORICAL RECORD of SOP

Rev# / Date	Written/Revised By	Revision Description	Approval
0. Written	Fanney Thorsteinsdottir		
1. Edited	Sabrina Wielsøe	Sens have updated the data base so it is possible to collect data on sleep, the setting for this is applied in the SOP.	Sofus Larsen 10.10.2024
2. Edited	Fanney Thorsteinsdottir	Data should be verified/cleaned when data is downloaded.	
3. Edited	Mie Juul Bække Petersen	Updated website instructions with new screenshots to match the updated Sens website	
4.			
5.			
6.			
Removed / replaced:			